



up to three days, medium range of three to seven days, extended range, beyond seven and up to 30 days and even for an entire season. But most often the forecast about the rains, beyond the short range, remains a gamble. This is because the IMD uses a set of historically determined atmospheric variables with significant statistical correlations to the total rainfall received by the country as a whole during June-September.

Tyagi, who has a doctoral degree in numerical weather prediction, has tried to incorporate new research trends to issue reliable forecasts. IMD weather analysts have already made a new beginning since 2009, using a dynamic forecasting system similar to that used by the advanced weather organisations in the world and, particularly, that of the European Centre for Medium-Range Weather Forecasts (ECMWF). Meanwhile, IMD is also making some changes in the frequency of forecasts, all to be made on Fridays, for monsoon 2010 (see box).

Even as meteorologists try to get a better grip on forecasting monsoons, comprehensive climate model simulations with respect to the potential of abrupt monsoon transitions are being studied by applying it to climate records of the past to

Friday FORECASTS

The IMD is all set to revise the frequency of its forecasts

Like previous years, this year also the first stage forecast of monsoon rainfall for June to September, 2010, was issued in April. Forecast for the onset of monsoon in Kerala will be issued in May. The second stage forecast of monsoon will be issued on June 25 which will cover the seasonal (June-September) forecast update, forecast for July and August rainfall for the country and forecasts for seasonal rainfall for four broad geographical regions—North-west India, North-east India, Central India and south peninsula. The mid-season update for the seasonal forecast and the forecast for August will be issued on July 30. The final update of the seasonal forecast and the forecast for September will be issued on August 27. IMD will also issue quantitative forecast for one week and outlook for the following week every Friday during the monsoon season. In effect, all forecasts will be out on Fridays.



“Long-term forecasts need to be revalidated as time goes by. Atmospheric science is dynamic, one cannot predict the future at once.”

ANUJ KUMBHAT, CO-FOUNDER, WEATHER RISK MANAGEMENT SERVICES

understand monsoon behaviour at the Indian Institute of Tropical Meteorology (IITM), Pune, and the Indian Institute of Science, Bangalore. “We are also making improvements in the forecast of the active and break cycle of the monsoon with a new statistical model developed by the IITM from this year,” explains Tyagi.

The Ministry of Earth Sciences has now proposed a national mission on the monsoon with the goal of developing reliable dynamic models by networking institutions and researchers. The aim is to offer credible three-day and five-day forecasts in about five years. With private forecasters also joining in, the competition is bound to improve the forecasts. Among them is Foreca which currently makes probabilistic forecasts for a maximum of 15 days ahead. “These are based on ensemble modelling done at ECMWF. In probabilistic forecasts, no attempt is made to predict the exact timing of rain, but merely the chance of rain occurring on any given day,” says Vishal Singh, country manager, Foreca.

Evidently, change is a constant in monsoon dynamics which is getting even more complicated. The future evolution will depend on global emissions of greenhouse gases as well as air pollution through aerosols—smoke in its various forms. All this makes forecasting monsoon rainfall for the future a daunting challenge. But it does not take one little raindrop away from the compelling need to get it right.

IMD's Operational Weather FORECASTING SYSTEM

OBSERVATIONS Data is collected from many sources

Surface Observations Manned/ Automatic (558 Stations)	Upper Air Observations (62 Stations)	Aircraft Reports	Ship Reports	Ocean Buoys Data	Satellite Observations	Global Data
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COMMUNICATION Information is then fed into a central hub 24x7

Global Telecom Hub/ Regional AMSS (data flow at three-hourly intervals on 24x7 mode)

ANALYSIS AND FORECASTING Analysts use the information to draw inferences

Delhi Northern Hemispheric Analysis Centre	Pune Weather Central	IMD, Delhi IAFS NWP forecasts
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DISSEMINATION Information is then disseminated and shared with other agencies

Agriculture, Aviation, Hydrology, Marine, Power, Defence, Public Warnings, Tourism